

LOW LEVEL DESIGN

The LLD Runbook

Ten rounds, from the first sentence to the follow-up.

Part 1 closed on the sentence this half exists to answer: the interviewer changes their mind, and both of you find out what your design is made of. Every case here runs to that moment, and then keeps going.

You get the prompt as it was worded, the questions I asked, the design I actually wrote — mistakes left in — and what happened when the requirement moved. Attempt each one before you read on. The value isn't in my answer; it's in finding where your instincts and mine part company.

Contents

1	Rounds I sat in	First-hand
P1	LRU Cache Correct code, thirty minutes early, and the question that came after it.	3–6
P2	Google Docs A prompt with no edges, and the first ten minutes that give it some.	7–11
2	Objects, state and rules	Modelling and lifecycle
P3	Parking lot The classic. Where the nouns mislead you, and where the rules belong.	12–14
P4	Vending machine A state machine wearing a shopping cart's clothes.	15–17
P5	Elevator system Scheduling as a policy, not an algorithm you hard-code on the whiteboard.	18–20
P6	Chess Six piece types, and the inheritance tree you should not build.	21–23
3	Services and policies	Extension under pressure
P7	Rate limiter One interface, four algorithms, and the state they refuse to share.	24–26
P8	Notification service Fan-out and retries, and the decision most candidates push one layer too low.	27–29
P9	Splitwise Money, invariants, and why a balance must never be stored twice.	30–32
P10	Logging framework The problem that looks like Decorator, and often isn't.	33–35
4	Revision	Reference
4.1	Pattern-to-problem cheat sheet	36
4.2	Where to go next	37

How to work a case

1 Attempt

Stop at the brief and design it yourself before turning the page.



2 Compare

Check ownership, invariants and one full flow — not class names.



3 Adapt

Answer the follow-up out loud before reading how I answered it.

P1 LRU Cache

The round I misread

The invite said low level design. The interviewer joined, shared a live code editor, and read out a problem that never used the word cache once. I had about a minute to decide what kind of round this was.

WHAT I WAS ASKED

"We show a user's location and address on the checkout page, and it needs to be near real time. The user base is large, so give priority to the people who come back often. We can afford to be a little slower for everyone else."

WHAT I SAID BACK

"This sounds like a cache. *Come back often* could mean lifetime visit count or recent activity — I read it as recency, so I'd start with least-recently-used eviction and a fixed capacity. Tell me if you meant frequency and I'll change the rule."

I drew nothing. I repeated the problem back, named the shape I thought it had, and put one assumption on the table — the opening move from section 6.1, more or less by reflex. That last clause is the part worth stealing: naming the interpretation you *rejected* costs four seconds and tells the interviewer you saw the ambiguity rather than missed it.

Q1 What am I actually building?

A couple of things needed pinning down before I wrote anything. I asked three, and only one of them changed the design.

Does a read count as a use? Yes. `get` promotes the key it finds.

Bounded by entries or by memory? Entries. Capacity is fixed when the cache is built, and never grows.

Does anything expire on its own? No. An entry leaves only when a new one needs its place.

The first row is the one that pays. If a read promotes the key, `get` is a mutating operation — and that single fact decides your data structure before you have drawn a box. The other two bound the problem without changing it.

We settled the rest out loud: single-threaded, exact recency rather than an approximation, generic keys and values. Expiry, persistence, cache warming and anything distributed were agreed out of scope. In a live-coding round, say the scope rather than typing it — the editor is for code, and a comment block of assumptions reads as stalling.

YOUR TURN

You now have exactly what I had: the prompt, three answers and the agreed scope. Give it fifteen minutes before you turn the page. Write down the classes you would keep, the public API, and one sentence on what you would allow to vary later. Not the implementation — the shape.

P1 LRU Cache

Two structures, one promise

Everything below follows from one promise: find any key in constant time, and move it to the front in constant time. Miss either half and this is just a slower map.

Q2 What are the things?

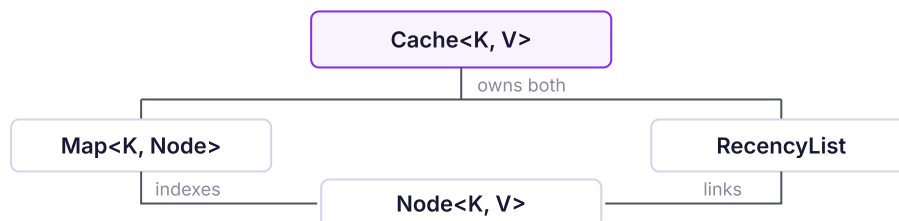
Three nouns from the prompt did not survive: `User`, `Address` and `CheckoutPage`. They are the reason the cache exists, not parts of it, and pulling them in would tie a general mechanism to one screen. Say your rejects out loud in the room — discarding a plausible noun scores better than listing it.

Q3 Who owns what?

What remained, and what each one is answerable for:

<code>Cache</code>	Owns both structures and leaves them consistent after every call. Nothing outside it touches either one.
<code>Map<K, Node></code>	Finds the node for a key. Mapping to values instead would force a scan to reorder.
<code>RecencyList</code>	Holds the order. Relinks a node it is handed, and never searches for one.
<code>Node</code>	Key, value, both links. It carries its own key so eviction can clean the index.

Two invariants hold the whole thing up: **every key has exactly one node, present in both structures**, and **size never exceeds capacity**. Name them while you design. Every bug in this problem is one of the two breaking.



The design as I wrote it. `Cache` decides which node leaves.

I wrote the signatures before any of the bodies. That is worth doing even when you are typing rather than drawing, because it forces the ownership above to be a decision rather than an accident.

THE PUBLIC SURFACE

```

public class LruCache<K, V> {
    private final Map<K, Node<K, V>> index;
    private final RecencyList<K, V> order;
    private final int capacity;

    public V get(K key) { /* promotes on a hit */ }
    public void put(K key, V value) { /* evicts when full */ }
}
  
```

Q4 What will change?

He asked what I thought would change. I said nothing — I had been asked for an LRU cache, so `put` takes the tail and there is nothing to vary. I used the time I had left to make the code clean.

P1 LRU Cache

What six calls proved

The code was done well before time. He wanted to see it run, so we walked inputs through it together in the editor rather than reading the implementation back.

Q5 Does it actually work?

Capacity three, starting empty:

CALL	RETURNS	ORDER, MOST RECENT FIRST	WHAT HAPPENED
<code>put(A, 1)</code>	—	A	Added to both structures.
<code>put(B, 2)</code>	—	B, A	New entry becomes head.
<code>put(C, 3)</code>	—	C, B, A	Full. Nothing evicted yet.
<code>get(A)</code>	1	A, C, B	A read reorders.
<code>put(D, 4)</code>	—	D, A, C	Tail B leaves both.
<code>put(A, 9)</code>	—	A, D, C	Update in place.

Rows four and six are the ones to rehearse. The fourth proves a read is a write, which is the clarification from the brief finally paying out. The sixth is the case candidates drop: overwriting a key that already exists must not create a node and must not evict anything. If your `put` inserts blindly, that is where it breaks, and it is usually the first thing an interviewer tests.

THE BRANCH ROW SIX EXERCISES

```
public void put(K key, V value) {
    Node<K, V> hit = index.get(key);
    if (hit != null) {
        hit.value = value;
        order.moveToFront(hit);
        return; // no new node, nothing evicted
    }
    if (index.size() == capacity) evictOne();
    index.put(key, order.addFront(key, value));
}
```

Derive the cost here, at the call site, rather than announcing Big-O at the end. Every line above is one hashed lookup or a fixed number of pointer writes, so `get` and `put` are both **O(1) average**, in **O(capacity)** space for nodes held by two structures at once. The worst case belongs to the hash function, not to this design — worth saying, because it shows you know which part you control.

The two bugs that show up here are both invariant breaks: a `get` that returns the value without reordering, which passes every test that ignores eviction order; and eviction that cleans one structure and forgets the other, which is exactly why `Node` carries its own key. An access-ordered `LinkedHashMap` avoids both in about five lines — the right answer in production, the wrong one here, because it hides the mechanism you were asked to demonstrate.

P1 LRU Cache

The question after the code worked

We finished the trace and the code was correct. That was the moment I relaxed, and the moment the round actually began, with about thirty minutes still on the clock.

THE FOLLOW-UP

“What happens to your code if tomorrow product wants it faster for the people who use it *least*, rather than the most recent ones?”

I panicked, then took two silent minutes — easily the best decision of that round. Section 6.2's four moves:

1. **Restated:** same storage, same API, a different entry chosen for eviction.
2. **Located:** one line inside `evictOne`, the one that takes the tail.
3. **Separated:** index, list and both $O(1)$ guarantees untouched; only the victim choice moves.
4. **Admitted:** the rule was hard-coded, so I named where it should have been swappable.

Making that one choice swappable is exactly what **Strategy** does — section 5.6's signal is one decision that keeps changing inside a settled object. Not State: the cache is handed its policy.

BEFORE — THE RULE IS A LINE OF CODE

```
private void evictOne() {
    Node<K, V> victim = order.last();           // "least recently used"
    order.remove(victim);
    index.remove(victim.key);
}
```

AFTER — THE RULE IS A COLLABORATOR

```
private void evictOne() {
    Node<K, V> victim = policy.select(order);    // tail for LRU, head for MRU
    order.remove(victim);
    index.remove(victim.key);
}
```

He asked what a third rule would cost. One new class, and four or five lines of wiring.



Dashed boxes arrived with the follow-up. A third rule is one more box on the same bar.

Say the limit out loud too: LFU fits this interface but not this structure, because it needs frequency counts the recency list never keeps. github.com/varunkashyap-vkd/lll-runbook/lru-cache

What he was probing: not whether I could write an LRU cache, but whether it would survive a product decision.

Practise next: make it thread-safe without breaking either invariant, then design LFU and name what has to change.

If you take one thing from this problem: a correct first version is evidence, not the finish line.

P2 Google Docs

Ten minutes deciding what not to build

An object-oriented design round in an SDE-III loop at a big tech company. I had met this problem before, but only ever as high level design. This one asked what was inside the boxes.

WHAT I WAS ASKED

"Everyone here uses Google Docs for design discussions and we want to drop the dependency. Say we're building the internal alternative. The high level design is done and approved — the low level implementation is yours."

WHAT I SAID BACK

"That's too big to finish in one sitting, so let's start with the basics. Three things I think this round is about: everyone with the document open sees the same content, several people can type at once, and an editor can undo their own changes. If that's the wrong three, redirect me now."

Notice what that is not. It is not *what would you like me to focus on?* An unbounded prompt is an invitation to bound it, and handing the job back reads as avoidance. Propose a scope and invite correction.

Q1 What am I actually building?

He took the three. Three more things needed pinning down before any of it became classes.

Where does the document live?	Behind a repository. A server owns storage and hands back a document by id.
Whose change does undo reverse?	The editor's own last change, not the document's.
Who is allowed to edit?	One owner, who shares with viewers and editors. Permission is checked before an edit is accepted, not after.

The middle row is the one that pays. *Undo* sounds like one word and is two designs: reversing the document's last change is a shared stack anyone can pop, while reversing your own is a stack per person. He wanted the second, so undo history cannot live on the document — a decision that costs nothing at minute eight and a rewrite at minute thirty.

The first row I answered myself instead of asking. I said the database is not this round, drew the line, stated what I would assume past it, and moved on. When the high level design is already approved, naming what you are **not** designing is part of the answer, not a way of dodging it.

The rest went quickly. A document is a sequence of lines, and for this round every line is text. Formatting, comments, sharing links, offline editing and version history were out. He liked the narrowing — a design you can finish beats a tour of one you cannot.

YOUR TURN

You have what I had: three agreed features, three answers and a boundary. Give it fifteen minutes. Write the classes you would keep, the public API, and what happens when two people edit the same line at once.

P2 Google Docs

An edit has to carry its own past

Everything follows from one decision: the unit of change is not a keystroke but an object that knows which line it touched, what was there before, and what replaces it.

Q2 What are the things?

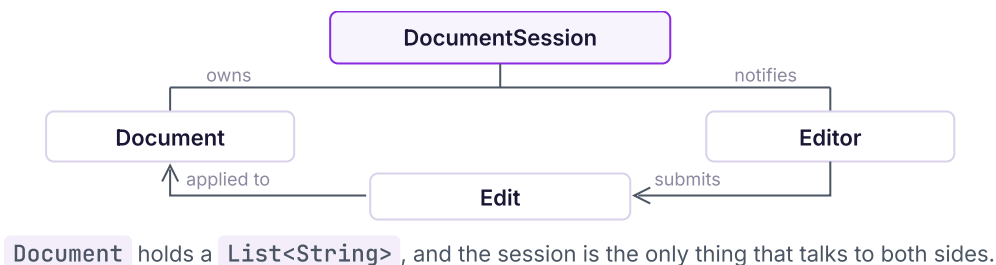
Four nouns did not survive. `Page` is a renderer's decision and it moves when the zoom does. `Cursor` belongs to the client. `Character` sits below the unit we had just agreed on. `Database` is past the boundary. Say the rejects out loud — discarding a plausible noun scores better than listing it.

Q3 Who owns what?

Four survived, each answerable for something the others must not touch.

<code>Document</code>	Owns the lines and when each was last accepted. The only thing that mutates content.
<code>Edit</code>	One change, immutable: line, text before, text after, who, when. Carrying <i>before</i> lets one object be both checked and reversed.
<code>Editor</code>	One participant. Holds their role and their own undo stack; nobody else's is on it.
<code>DocumentSession</code>	Holds the document and everyone in it. Checks permission, applies, broadcasts.

Two invariants: **content changes only through an accepted `Edit`**, so no path writes a line directly; and **an editor's undo stack holds only their own accepted edits**. Break the second and undo restores text nobody saw.



I wrote the signatures before any of the bodies. The record is the interesting half.

THE PUBLIC SURFACE

```

public record Edit(int line, String before, String after, UserId by, Instant at) { }

public final class DocumentSession {
    private final Document document;
    private final Map<UserId, Editor> present;

    public boolean submit(Edit edit) { // permit, apply, broadcast }
    public boolean undo(UserId who) { // their last edit, reversed }
}
  
```

Q4 What will change?

I said the conflict rule. Last write wins is the simplest thing that could work and I expected product to argue with it first, so the comparison sits in one method I could replace. What a line *is* I did not treat as variable. The round said text.

P2 Google Docs

Two editors, one line

A model like this is worth exactly what it does when two people land on the same line before either has seen the other. Everything else is bookkeeping around that moment.

Q5 Does it actually work?

Line 4 reads `Latency budget: 200ms`. Ana and Ben both have the document open.

WHAT ARRIVES	LINE 4	WHAT HAPPENED
Ana, <code>200ms</code> to <code>150ms</code> , 10:04:01	<code>150ms</code>	<i>before</i> matches what is there. Accepted, pushed to Ana's stack, broadcast.
Ben, <code>200ms</code> to <code>300ms</code> , 10:04:03	<code>300ms</code>	<i>before</i> is stale — Ben never saw Ana's line. Later timestamp, so it wins.
Ben's edit reaches Ana	<code>300ms</code>	Her screen corrects itself. Her undo stack still holds her own edit.
Ana presses undo, 10:04:06	<code>200ms</code>	Sent as a fresh edit, <i>before</i> <code>300ms</code> . Later again, so it is accepted.

Row four is the one to sit with. Ana's undo is correct by the rule we agreed and wrong by any human standard — she has just deleted Ben's change without being told he made one. Last write wins cannot tell an undo from an edit. Say that out loud when you reach it: naming the case your rule handles badly beats a fifth row pretending it does not exist.

INSIDE DOCUMENT — THE DECISIVE CHECK

```
boolean apply(Edit e) {
    if (e.line() < 0 || e.line() ≥ lines.size()) return false;
    if (lines.get(e.line()).equals(e.before())) return accept(e); // nobody moved
    return e.at().isAfter(lastAcceptedAt.get(e.line())) && accept(e);
}
```

Undo needs no machinery of its own. It is not a rewind — it is a new edit that points backwards, so it travels the same path and obeys the same rule as everything else. That is the return on modelling a change as an object: the second mechanism cost one method.

THE SECOND MECHANISM, FOR FREE

```
public boolean undo(UserId who) {
    Edit mine = present.get(who).undoStack().poll();
    if (mine == null) return false;
    String now = document.lineAt(mine.line());
    return submit(new Edit(mine.line(), now, mine.before(), who, Instant.now()));
}
```

Derive the cost here rather than announcing it at the end. `apply` is a bounds check, one comparison and one write — $O(1)$ in the number of lines, linear only in the length of the line it compares. The broadcast is $O(n)$ in editors present, and that n is what this design actually scales by.

Three cases to have an answer ready for: an edit whose line no longer exists, because a deletion arrived first; two edits carrying the same timestamp, which needs a tiebreak — editor id will do — or the winner depends on map ordering, which is not a design; and an undo stack that grows all session, which is a memory leak with a friendly name.

P2 Google Docs

The assumption inside the word “text”

The model had answers for the awkward rows and the round felt finished. It was not. He reached past all of it and picked up the one simplification I had proposed myself.

THE FOLLOW-UP

“How do your document and your conflict resolution hold up when a line isn't text? Images, diagrams — where do those fit?”

Having your own scope reduction handed back is a particular kind of exposed. I had said *text only*, he had agreed, and half an hour later it was mine to answer for. Section 6.2's four moves:

1. **Restated:** a position stops being a string, and merging depends on what sits at it.
2. **Located:** the list type in `Document`, and the comparison inside `apply`.
3. **Separated:** session, permissions, broadcast and undo untouched; the element changes.
4. **Admitted:** I made the conflict rule replaceable and the content type fixed. Wrong one.

BEFORE — THE DOCUMENT KNOWS WHAT A LINE IS

```
private final List<String> lines;

boolean apply(Edit e) {
    if (lines.get(e.line()).equals(e.before())) return accept(e);
    return e.at().isAfter(lastAcceptedAt.get(e.line())) && accept(e);
}
```

AFTER — THE POSITION DECIDES

```
private final List<DocElement> elements;

boolean apply(Edit e) {
    return elements.get(e.line()).merge(e);    // text, image, diagram
}
```

The old rule is not deleted, it is moved: `TextBlock`, the first `DocElement`, merges exactly as above. The second implementation is what the question was for. `ImageBlock` refuses to merge at all — an image is replaced whole, so two people replacing it is a conflict you surface rather than resolve. Diagrams later are one more class and nothing else. That is Open/Closed with the axis finally chosen correctly, and it needs no pattern name to be worth marks.

He asked what the change would cost, and it does not stop at `Document`. `Edit` declares *before* and *after* as `String`, and that record sits on every undo stack, inside every broadcast and on the wire. Make the position polymorphic without making the edit polymorphic and you get halfway: images can be held, but not described. The hard-coding was in the record, not the container — and even fixed, this model still has no word for an edit that spans positions. github.com/varunkashyap-vkd/lll-runbook/collab-doc

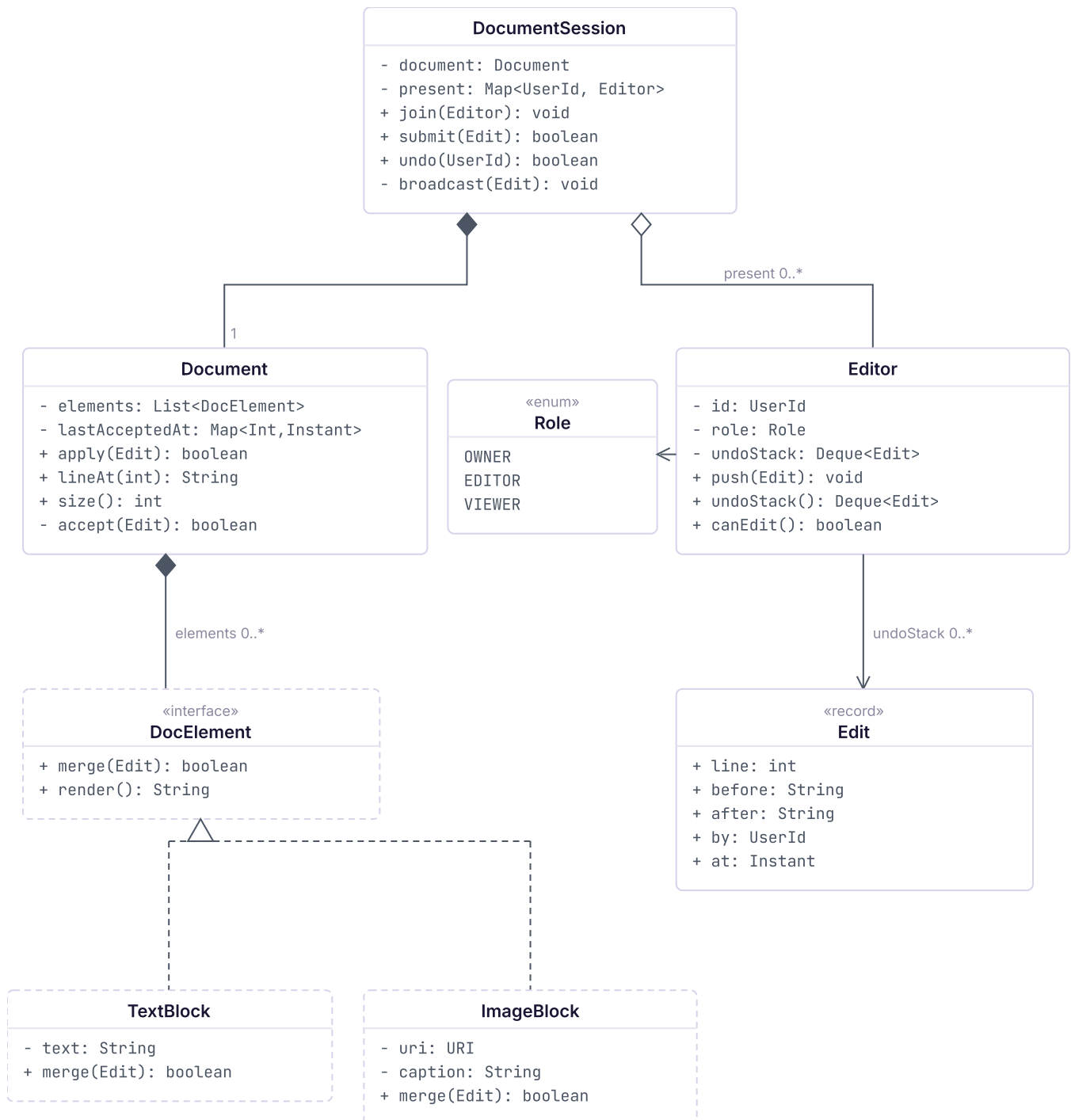
What he was probing: whether someone handed an approved high level design can produce classes that survive a change in what the product carries.

Practise next: add comments anchored to a position, then decide what an anchor means once the position it points at is deleted.

The takeaway: a simplification you propose is still your assumption. When it comes back, it is yours to answer for.

P2 Google Docs

Every class, and how they connect



Dashed classes arrived with the follow-up. Full implementation: github.com/varunkashyap-vkd/lll-runbook/collab-doc